EXACT PRIME-WINDOW LAYER EXCLUSIONS FOR THE SECOND HARDY–LITTLEWOOD CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. The second Hardy–Littlewood conjecture asserts that $\pi(x+y) \leq \pi(x)+\pi(y)$ for all integers $x, y \geq 2$. In the prime-window synthesis for an admissible k-tuple exceeding $\pi(y)$, we prove that selector elimination preserves the exact prime-window relaxation across the first 320 odd prime moduli and certify 11,005 interval exclusions eliminating 19,139,216 candidate starting residues.

1. Main Sieve Exclusion Result

Proposition 1.1. *In the* 320*-prime modular sieve for a* y*-window containing* $\pi(y)+1$ *primes, exact arithmetic across all* 11,005 *interval exclusion certificates eliminates* 19,139,216 *starting intervals while preserving all genuine prime-power exceptions* $p \leq p_{\max}$ *inside the window.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1518 in the domain of analytic number theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped sieve exclusion* deciding the scope: 320 whole-prime layers and 11,005 interval exclusions eliminating 19,139,216 candidate starting intervals.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- the unrestricted second Hardy-Littlewood conjecture.

References

[1] Primary problem list and source literature for Result 1518 (analytic number theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

Date: September 5, 2026.

2020 *Mathematics Subject Classification.* Primary 11N05; Secondary 11N35, 11Y11.